

25.3 Confounding and causal graphs

The functions and disturbances of a structural causal model are usually unknown; its parent sets, saying which variables can influence which, are qualitative and best displayed as a graph. This section shows how the graph constrains the joint distribution and identifies confounding as a property of paths. For linear models the connection is exact, and the omitted-variable bias of Proposition 5.7.2 becomes a sum over paths.

25.3.1 Graphs

Definition 25.3.1. Directed acyclic graph

A **directed graph** on nodes $1, \dots, m$ is a set of ordered pairs (l, j) , $l \neq j$, drawn as arrows $l \rightarrow j$. It is **acyclic** (a DAG) if no sequence of arrows leads from a node back to itself. In a DAG:

- (a) l is a **parent** of j , and j a **child** of l , if $l \rightarrow j$;
- (b) a **path** between l and j is a sequence of distinct nodes, starting at l and ending at j , in which consecutive nodes are joined by an arrow in either direction;
- (c) a path is **directed** (from l to j) if every arrow on it points towards j ; then l is an **ancestor** of j , and j a **descendant** of l ;
- (d) an interior node c of a path is a **collider** on the path if both arrows of the path that meet at c point into it, $\rightarrow c \leftarrow$; every other interior node is a **non-collider** on the path.

The **graph of a structural causal model** has an arrow $l \rightarrow j$ iff $l \in \text{pa}(j)$. It is acyclic because parents have smaller indices.

Being a collider is a property of a node *on a given path*: in Figure 25.3.1(c), C is a collider on $X \rightarrow C \leftarrow Y$ but not on a path that continues along an arrow $C \rightarrow D$.

25.3.2 What the graph says about the distribution

Assume that for each j and each value $\mathbf{v}$ of the parents, the variable $f_j(\mathbf{v}, U_j)$ has a density $p_j(\cdot \mid \mathbf{v})$ with respect to Lebesgue or counting measure.

Proposition 25.3.1. Markov factorization and truncated factorization

Under this assumption:

- (a) $(V_1, \dots, V_m)$ has density

$$p(v_1, \dots, v_m) = \prod_{j=1}^m p_j(v_j \mid v_{\text{pa}(j)}), \quad (25.3.1)$$

and $p_j(\cdot \mid v_{\text{pa}(j)})$ is the conditional density of V_j given $V_1, \dots, V_{j-1}$, which depends on them only through the parents;

- (b) under $\text{do}(V_k = x)$, the other variables have density $\prod_{j \neq k} p_j(v_j \mid v_{\text{pa}(j)})$, evaluated with $v_k = x$.

Proof.

- (a) The variables $V_1, \dots, V_{j-1}$ are functions of $U_1, \dots, U_{j-1}$, which are independent of U_j . For independent inputs, the conditional distribution of $f_j(V_{\text{pa}(j)}, U_j)$ given $(V_1, \dots, V_{j-1}) = (v_1, \dots, v_{j-1})$ is the distribution of $f_j(v_{\text{pa}(j)}, U_j)$ (Billingsley 1995), whose density is $p_j(\cdot | v_{\text{pa}(j)})$. The factorization follows by multiplying these conditional densities in the order $j = 1, \dots, m$.
- (b) The modified model is a structural causal model for the variables other than V_k , in which V_k is the constant x and every other assignment and disturbance is unchanged. Apply (a) to it.

■

So an intervention deletes one factor and keeps the others, which are observational conditional densities; with every variable observed, the interventional distribution is computable from the observational one.

The linear case has an explicit form. Write a linear model as $\mathbf{V} = \mathbf{c} + \mathbf{B}\mathbf{V} + \mathbf{U}$, where $\mathbf{B} = (b_{jl})$ has $b_{jl} \neq 0$ only if $l \in \text{pa}(j)$, so $\mathbf{B}$ is strictly lower triangular, and $\mathbf{\Omega} = \text{Cov}(\mathbf{U}) = \text{diag}(\omega_1, \dots, \omega_m)$. The coefficient b_{jl} labels the arrow $l \rightarrow j$.

Proposition 25.3.2. Path tracing

In a linear structural causal model:

- (a) $\mathbf{I} - \mathbf{B}$ is invertible, $\mathbf{V} = \mathbf{A}(\mathbf{c} + \mathbf{U})$ with $\mathbf{A} = (\mathbf{I} - \mathbf{B})^{-1}$, and $\text{Cov}(\mathbf{V}) = \mathbf{A}\mathbf{\Omega}\mathbf{A}^\top$;
- (b) for $j \neq l$, the entry a_{jl} of $\mathbf{A}$ is the sum, over all directed paths from l to j , of the products of the coefficients on the arrows of the path (zero if there is none), and $a_{jj} = 1$;
- (c) $E(V_j | \text{do}(V_l = x)) = E(V_j | \text{do}(V_l = 0)) + a_{jl}x$.

The number a_{jl} is the **total effect** of V_l on V_j : the change in the interventional mean of V_j per unit change in the value imposed on V_l .

Proof.

- (a) $\mathbf{I} - \mathbf{B}$ is unit lower triangular, hence invertible, and $(\mathbf{I} - \mathbf{B})\mathbf{V} = \mathbf{c} + \mathbf{U}$. The covariance follows from Theorem 2.2.1.
- (b) $\mathbf{B}$ is strictly lower triangular, so $\mathbf{B}^m = \mathbf{0}$ and $(\mathbf{I} - \mathbf{B})(\mathbf{I} + \mathbf{B} + \dots + \mathbf{B}^{m-1}) = \mathbf{I} - \mathbf{B}^m = \mathbf{I}$. The (j, l) entry of $\mathbf{B}^r$ is $\sum b_{jk_1} b_{k_1 k_2} \dots b_{k_{r-1} l}$ over all $k_1, \dots, k_{r-1}$; a term is nonzero only if $l \rightarrow k_{r-1} \rightarrow \dots \rightarrow k_1 \rightarrow j$ are arrows, which is a directed path of length r (its nodes are distinct because the graph is acyclic). Summing over r gives (b).
- (c) The intervened model is linear with $\mathbf{B}$ replaced by $\tilde{\mathbf{B}}$, in which row l is zero, and with $c_l + U_l$ replaced by x . A directed path out of l never uses an arrow into l , so the directed paths from l in the graph of $\tilde{\mathbf{B}}$ are those of $\mathbf{B}$, and by (b) the (j, l) entry of $(\mathbf{I} - \tilde{\mathbf{B}})^{-1}$ is a_{jl} . By (a) applied to the intervened model, V_j is $a_{jl}x$ plus terms not involving x .

■

This is Wright's method of path coefficients (Wright 1921, 1934); the covariance formula in (a) gives his rule for correlations ([Exercise \(C1\)](#)).

25.3.3 Chains, forks and colliders

Three configurations of three variables are the building blocks of everything the graph says about conditional independence ([Figure 25.3.1](#)).

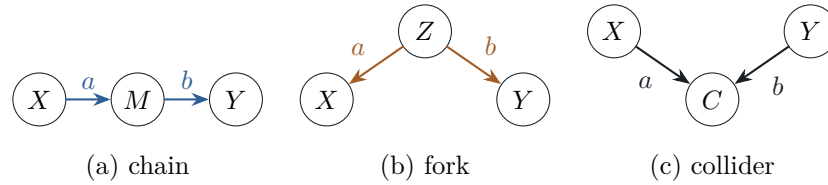


Figure 25.3.1. The three structures of [Proposition 25.3.3](#). In (a) and (b) the middle node transmits association between X and Y , and conditioning on it removes the association. In (c) the middle node transmits nothing until it is conditioned on.

Proposition 25.3.3. Chains, forks and colliders

Consider linear models with independent disturbances, $\text{Var } U_X = \sigma_X^2 > 0$ and so on, and write $\sigma_{XY \cdot W}$ for the partial covariance of X and Y given W ([Definition 3.6.1](#)).

- (a) **Chain** $X = U_X$, $M = aX + U_M$, $Y = bM + U_Y$: $\text{Cov}(X, Y) = ab\sigma_X^2$ and $\sigma_{XY \cdot M} = 0$.
- (b) **Fork** $Z = U_Z$, $X = aZ + U_X$, $Y = bZ + U_Y$: $\text{Cov}(X, Y) = ab\sigma_Z^2$ and $\sigma_{XY \cdot Z} = 0$.
- (c) **Collider** $X = U_X$, $Y = U_Y$, $C = aX + bY + U_C$: $\text{Cov}(X, Y) = 0$ and

$$\sigma_{XY \cdot C} = -\frac{ab\sigma_X^2\sigma_Y^2}{a^2\sigma_X^2 + b^2\sigma_Y^2 + \sigma_C^2}. \quad (25.3.2)$$

With normal disturbances, a zero partial covariance means conditional independence ([Proposition 3.6.1](#)), so in (a) and (b) X and Y are dependent but conditionally independent given the middle node, and in (c) they are independent but, if $ab \neq 0$, conditionally dependent given C .

Proof. The covariances follow from [Proposition 25.3.2\(a\)](#). (a) $Y - bM = U_Y$ is uncorrelated with X and M , so $L(Y \mid M, X) = bM$ has no X term, and by [Proposition 14.5.2](#) $\sigma_{XY \cdot M} = 0$. (b) The same argument with $Y - bZ = U_Y$. (c) By definition,

$$\sigma_{XY \cdot C} = \text{Cov}(X, Y) - \frac{\text{Cov}(X, C)\text{Cov}(Y, C)}{\text{Var}(C)} = 0 - \frac{(a\sigma_X^2)(b\sigma_Y^2)}{\text{Var}(C)}.$$

■

In the collider, independent causes become dependent once their common effect is held fixed. In the listing, with $a = b = 1$ and unit variances, the correlations of X and Y in the chain and the fork are 0.577 and 0.500 (simulated: 0.582 and 0.505), and both partial correlations given the middle node are zero. In the collider the correlation is zero and the partial correlation given C is -0.500 (simulated -0.501).

```
import numpy as np
```

```

rng = np.random.default_rng(2503)
n = 100_000
a, b = 1.0, 1.0                                # the two edge coefficients

def corr_and_partial(x, y, w):
    """Correlation of x and y, and their partial correlation given w (residuals on
    1, w)."""
    W = np.column_stack([np.ones(len(w)), w])
    rx = x - W @ np.linalg.lstsq(W, x, rcond=None)[0]
    ry = y - W @ np.linalg.lstsq(W, y, rcond=None)[0]
    return np.corrcoef(x, y)[0, 1], np.corrcoef(rx, ry)[0, 1]

U = rng.normal(size=(3, n))
x = U[0]; m = a * x + U[1]; y = b * m + U[2]      # chain    X -> M -> Y
chain = corr_and_partial(x, y, m)
z = U[0]; x = a * z + U[1]; y = b * z + U[2]      # fork      X <- Z -> Y
fork = corr_and_partial(x, y, z)
x = U[0]; y = U[1]; c = a * x + b * y + U[2]      # collider  X -> C <- Y
collider = corr_and_partial(x, y, c)
for name, (r, rp) in [("chain", chain), ("fork", fork), ("collider", collider)]:
    print(f"{name:9s} corr {r:6.3f}    partial corr given middle {rp:6.3f}")

```

[Run this code online](#)

25.3.4 d-separation

In a larger graph association can travel along many paths at once; d-separation combines the three structures.

Definition 25.3.2. d-separation

Let $\mathcal{Z}$ be a set of nodes. A path is **blocked** by $\mathcal{Z}$ if

- (i) some non-collider on the path is in $\mathcal{Z}$, or
- (ii) some collider on the path is not in $\mathcal{Z}$ and has no descendant in $\mathcal{Z}$.

Otherwise the path is **open** given $\mathcal{Z}$. Disjoint sets of nodes $\mathcal{A}$ and $\mathcal{B}$ are **d-separated** by $\mathcal{Z}$ (disjoint from both) if every path between a node of $\mathcal{A}$ and a node of $\mathcal{B}$ is blocked by $\mathcal{Z}$.

A path is open when each non-collider is left free and each collider is conditioned on, directly or through a descendant, a noisy measurement of it ([Exercise \(B1\)](#)).

Theorem 25.3.4. d-separation and conditional independence

Let $\mathbf{V}$ follow a structural causal model with graph G , and let $\mathcal{A}, \mathcal{B}, \mathcal{Z}$ be disjoint sets of nodes.

- (a) If $\mathcal{Z}$ d-separates $\mathcal{A}$ and $\mathcal{B}$ in G , then $\mathbf{V}_{\mathcal{A}}$ and $\mathbf{V}_{\mathcal{B}}$ are conditionally independent given $\mathbf{V}_{\mathcal{Z}}$.
- (b) If $\mathcal{Z}$ does not d-separate them, there is a linear structural causal model on G with normal disturbances in which they are conditionally dependent; in fact this holds for all edge coefficients and disturbance variances outside a set of Lebesgue measure zero.

Proof. Sketch; full proofs are in the references. For (a) with densities (Lauritzen, Dawid, Larsen and Leimer 1990): the nodes of $\mathcal{A} \cup \mathcal{B} \cup \mathcal{Z}$ and their ancestors form a set $\mathcal{N}$ whose density, by Proposition 25.3.1, is a product of functions of the sets $\{j\} \cup \text{pa}(j)$, $j \in \mathcal{N}$. These sets are complete in the *moral graph* on $\mathcal{N}$, which joins each node to its parents and any two parents of a common child. If $\mathcal{Z}$ separates $\mathcal{A}$ from $\mathcal{B}$ there, the factors split into two groups, giving the conditional independence; a combinatorial lemma shows that d-separation in G is equivalent to this separation. Geiger, Verma and Pearl (1990) give another proof. For (b), partial correlations of a linear normal model are rational functions of the parameters, not identically zero along an open path, and so vanish only on a set of measure zero (Spirtes, Glymour and Scheines 2000). ■

Part (b) says the converse of (a) fails only by accident, as in Exercise (A1), where the effect of visits and the association through illness cancel exactly. Distributions with no such cancellation are called **faithful** to the graph.

25.3.5 Confounding as an open path

Definition 25.3.3. Back-door paths and confounding

A **back-door path** from X to Y is a path between them whose arrow at X points into X , $X \leftarrow \cdots Y$. The effect of X on Y is **confounded** if some back-door path from X to Y is open given the empty set, that is, contains no collider.

Directed paths from X to Y carry the effect of X ; back-door paths carry association that reflects whatever caused X . In a linear model the split is exact.

Proposition 25.3.5. Confounding bias in a linear model

In a linear structural causal model with disturbances of positive variance, let $X = V_k$, $Y = V_j$, and let $\tau = a_{jk}$ be the total effect of X on Y . For each l , let d_l be the sum over the directed paths from V_l to Y that do not pass through X of the products of coefficients (so $d_k = 0$, and $d_j = 1$ from the trivial path of Y to itself).

- (a) $Y = \tau X + R$, where $R = \sum_l d_l(c_l + U_l)$ is the value of Y under $\text{do}(X = 0)$. Under $\text{do}(X = x)$, $Y = \tau x + R$.
- (b) The slope of the best linear predictor $L(Y | X)$ is

$$\tau + \frac{\text{Cov}(R, X)}{\text{Var}(X)}, \quad \text{Cov}(R, X) = \sum_l \omega_l d_l a_{kl}. \quad (25.3.3)$$

- (c) If the effect of X on Y is not confounded, the slope of $L(Y | X)$ is τ .

Proof.

- (a) A directed path from V_l to Y either avoids X or passes through it exactly once, and a path of the second kind splits at X into a directed path from V_l to X and one from X to Y . Summing products over paths with Proposition 25.3.2(b), $a_{jl} = d_l + a_{kl}\tau$ for

every l (for $l = k$, $a_{kk} = 1$ and $d_k = 0$; for $l = j$, $a_{jj} = 1 = d_j$ and $a_{kj} = 0$). Hence $Y = \sum_l a_{jl}(c_l + U_l) = \tau \sum_l a_{kl}(c_l + U_l) + R = \tau X + R$. Under $\text{do}(X = x)$, as in the proof of Proposition 25.3.2(c), the arrows into X are removed, so the directed paths from V_l , $l \neq k$, to Y in the intervened graph are exactly those that avoid X . Thus $Y = \tau x + \sum_{l \neq k} d_l(c_l + U_l) = \tau x + R$.

- (b) The slope is $\text{Cov}(X, Y) / \text{Var}(X) = \tau + \text{Cov}(R, X) / \text{Var}(X)$ by Theorem 6.11.1, and $\text{Cov}(R, X) = \sum_{l, l'} d_l a_{kl'} \text{Cov}(U_l, U_{l'}) = \sum_l \omega_l d_l a_{kl}$, since the disturbances are uncorrelated.
- (c) Suppose $\omega_l d_l a_{kl} \neq 0$ for some l . Then $l \neq k$, and there are a directed path π_1 from V_l to X and a directed path π_2 from V_l to Y that avoids X . Let w be the node of π_1 closest to X that also lies on π_2 ; it exists because V_l lies on both, and $w \neq X$ because π_2 avoids X . Follow π_1 backwards from X to w , then π_2 forwards from w to Y . By the choice of w the nodes are distinct, so this is a path. Its arrow at X is the last arrow of π_1 , which points into X , and every arrow on it points away from w , so it has no collider. The effect is therefore confounded. Contrapositively, if it is not confounded, every term of $\text{Cov}(R, X)$ vanishes.

■

Part (a) is the potential outcome $Y(x) = \tau x + Y(0)$: linear structural models have constant effects. Part (b) writes the regression slope as the effect plus one contribution for each common ancestor of X and Y . In the model of Figure 25.1.1, $R = c_Y + \gamma Z + U_Y$ and the only term comes from Z : $\omega da = \sigma_Z^2 \cdot \gamma \cdot \alpha$, which is Eq. 25.1.3.

This is the omitted-variable bias of least squares: by Proposition 5.7.2 the short regression estimates $\theta + \pi\gamma$, with π the sample slope of $\mathbf{z}$ on $\mathbf{x}$, whose limit is $\alpha\sigma_Z^2 / \text{Var}(X)$ (see also Theorem 19.1.2 in Chapter 19). The graph adds that an omitted variable causes bias only if it lies on an open back-door path; an omitted mediator should stay omitted (Section 25.5).

25.3.6 Exercises

25.3.6.1 A. Check your understanding

Exercise. A1

A DAG has arrows $A \rightarrow X$, $A \rightarrow B$, $B \rightarrow Y$, $X \rightarrow Y$, $X \rightarrow C$, $Y \rightarrow C$ and $D \rightarrow C$. List every path between X and Y , mark the back-door paths, and say which paths are open given $\emptyset$, given $\{B\}$ and given $\{C\}$.

Solution. There are three paths: $X \rightarrow Y$; $X \leftarrow A \rightarrow B \rightarrow Y$, a back-door path; and $X \rightarrow C \leftarrow Y$, with collider C . (The arrow $D \rightarrow C$ gives no path to Y except through C , where $D \rightarrow C \leftarrow Y$ would require revisiting C .) Given $\emptyset$: the first two are open and the third is blocked by the collider. Given $\{B\}$: only $X \rightarrow Y$ is open. Given $\{C\}$: all three are open, since conditioning on the collider opens the third.

25.3.6.2 B. Practice

Exercise. B1

Extend the collider by $D = C + U_D$, with $\text{Var } U_D = \delta$ and all other variances one. Compute $\sigma_{XY \cdot D}$ and show that it lies between 0 and $\sigma_{XY \cdot C}$, approaching them as $\delta \rightarrow \infty$ and $\delta \rightarrow 0$.

Solution. $\text{Cov}(X, D) = a$, $\text{Cov}(Y, D) = b$ and $\text{Var } D = a^2 + b^2 + 1 + \delta$, so $\sigma_{XY \cdot D} = -ab/(a^2 + b^2 + 1 + \delta)$. At $\delta = 0$ this is [Eq. 25.3.2](#), and it decreases in absolute value to zero as δ grows. A proxy for a collider opens the path partially, the more so the better the proxy.

25.3.6.3 C. Going deeper

Exercise. C1

Show from [Proposition 25.3.2\(a\)](#) that in a linear model $\text{Cov}(V_j, V_k) = \sum_l \omega_l a_{jl} a_{kl}$. Interpret each term as a sum over pairs of directed paths from V_l to V_j and to V_k , and use this to compute $\text{Cov}(X, Y)$ in the chain of [Proposition 25.3.3](#).